

% Normalization of Alumni Network and Engagement Platform %

• Department :

Candidate key : DeptID

Functional Dependency : ~~DeptID~~ → ~~DeptName~~,
DeptCode

→ Department (DeptID, DeptName, DeptCode, HODName, Established Year)

DeptID → { DeptName, DeptCode, HODName, EstablishedYear }

DeptName → DeptID (candidate key)

DeptCode → DeptID (candidate key)

DeptID	DeptName	DeptCode	HOD NAME	Established Y
1	Computer Science	CSE	Dr. R. Sharma	1998
2	Electronics & Comm	ECE	Dr. S. Iyer	1999
3	Mechanical Engg.	MECH	Dr. A. Verma	1995
4	Electrical Engg.	EEE	Dr. P. Nair	2001
5	Civil Engg.	CIVIL	Dr. K. Rao	1994

• Normal-form check

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single-attribute primary key - a partial dependency is impossible
3NF	✓	DeptName and DeptCode are no candidate keys, so there is no non-prime → non-prime dependency

BCNF



every determinant
(DeptID, DeptName, DeptCode) is
a candidate key.

•

Batch

Functional Dependencies :

BATCH-flat (BatchID, BatchYear, Section, Total Students, DeptID, DeptName, StudentID, StudentName)

BatchID $\rightarrow$ BatchYear, Section, Total Students, DeptID

DeptID $\rightarrow$ DeptName (transitive dependency)

StudentID $\rightarrow$ StudentName (partial dependency)

$\Rightarrow$ Candidate Key(s) : (BatchID, StudentID) -
composite key of the flat relation.

BatchID	BatchYear	Section	TotalStudents	DeptID
1	2023	A	120	1
2	2023	B	118	1
3	2024	A	125	2
4	2024	B	122	3
5	2025	A	130	4

Normal-Form Check:

Normal Form	Status	Reasoning
1NF	✓	After flattening, every cell holds one atomic value.
2NF	✓	Partial Dependency StudentID $\rightarrow$ StudentName removed by projection (Step 1)

3NF

✓

Transitive dependency
 $\text{BatchID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$
 removed by projection
 (step 2)

BCNF

✓

Determinants BatchID
 (in Batch) and studentID
 (in the student roster) are
 candidate keys.

• Decomposition:

Step 1: 2NF : remove partial dependency $\text{StudentID} \rightarrow \text{StudentName}$.

→ Project the relation over two dependencies

$R1 = \text{Batch facts} (\text{BatchID} \rightarrow \{\text{BatchYear}, \text{Section}, \text{TotalStudents}, \text{DeptID}\})$

BatchID	Batch Year	Section	TotalStudents	DeptID
1	2023	A	120	1
1	2023	A	120	1
3	2024	A	125	2
4	2024	B	122	3
5	2025	A	130	4

$R2 = \text{Student-Roster} (\text{StudentID} \rightarrow \text{StudentName})$

StudentID	Student Name	BatchID
S0001	Ravi Kumar	1
S0002	Anita Shah	1
S0003	Farhan Ali	3
S0004	Lakshmi Bhat	4
S0005	Mohan Das	5

Step 2: 3NF : remove the transitive dependency
 $\text{BatchID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

$\Rightarrow$ DeptName depends on DeptID, which depends on BatchID. Project DeptName; it merges into Department

Batch (final):

BatchID	BatchYear	Section	TotalStudents	DeptID (FK)
1	2023	A	120	1
1	2023	A	120	1
3	2024	A	125	2
4	2024	B	122	3
5	2025	A	130	4

(DeptID, DeptName) - merges into Department:

DeptID	DeptName
1	Computer Science
2	Electronics & Comm.
3	Mechanical Engg.
4	Electrical Engg.
5	Civil Engg.

• Company

Functional Dependencies:

Company (CompanyID, CompanyName, Industry, CompanySize, Website, Headquarters)

CompanyID $\rightarrow$ CompanyName, Industry, CompanySize, Website, Headquarters

Candidate Key : Company ID

Company ID	Company Name	Industry	Company Size	Website
101	TCS	IT Services	500000+	tcs.com
102	Infosys	IT Services	300000+	infosys.com
103	Google	Technology	100000+	google.com
104	Reliance	Conglomerate	200000+	reliance.com
105	Zoho	Software	15000	zoho.com

Headquarters

Mumbai

Bengaluru

Mountain View

Mumbai

Chennai

Normal Form Check:

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single attribute primary key - no partial dependency
3NF	✓	All non-key attributes depend directly on Company ID i.e. no transitive dependency
BCNF	✓	Company ID is the only determinant and is a candidate key.

• Skill

- SkillID is a single-attribute primary key. SkillName is also a candidate key. Every determinant is a candidate key, so the relation already in BCNF.

~~Initial~~ Functional Dependencies:

Skill (SkillID, SkillName, SkillCategory, Description)

SkillID → SkillName, SkillCategory, Description

SkillName → SkillID (candidate key)

Candidate Key: SkillID (SkillName is an alternative primary key)

SkillID	SkillName	SkillCategory
1	Java	Programming
2	Python	Programming
3	SQL	Database
4	Communication	Soft Skill
5	Project Management	Management

Description

COOP and backend development

Data Analysis and automation

Querying and database design

Verbal and written communication

Planning and Team coordination

Normal Form Check

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single - attribute primary key - no partial dependency
3NF	✓	No non-prime → non-prime dependency
BCNF	✓	SkillID and SkillName are both candidate keys; every determinant is a candidate key.

Alumni:

- AlumniID is a single-attribute ^{primary} key (Email is an alternative candidate key).
 Once the transitive dependencies are removed, every determinant is a candidate key → the relation is in BCNF

Functional Dependencies:

Alumni-flat (AlumniID, FirstName, LastName, Email, Date of Birth, Gender, Address-City, Address-State, Address-PinCode, Graduation Year, DeptID, DeptName, BatchID, Batch Year, CompanyID, Company Name, Current Position, LinkedIn Profile, IsActive)

AlumniID → { FirstName, LastName, Email, Date of Birth, Gender, Address-City, Address-State, Address-PinCode, Graduation Year, DeptID, BatchID, CompanyID, Current Position, LinkedIn Profile, IsActive }

Email → AlumniID (candidate key)
 DeptID → DeptName (transitive)
 BatchID → BatchYear (transitive)
 companyID → CompanyName (transitive)
 Candidate Key : AlumniID (Email is an
 alternative
 candidate key)

AlumniID	First Name	Last Name	Email	Date of Birth
1	Aarav	Mehra	aarav.mehra @email.com	2001-04-12
2	Pritya	Nair	pritya.nair@ email.com	2000-11-03
3	Rohan	Gupta	rohan.gupta@ email.com	2000-07-25
4	Sneha	Reddy	sneha.reddy@ email.com	2002-01-19
5	Vikram	Singh	vikram.singh @email.com	2002-09-30

Gender	Address-City	Address-State	Address-Pincode
M	Chennai	TN	600001
F	Bengaluru	KA	560001
M	Mumbai	MH	400001
F	Hyderabad	TS	500001
M	Delhi	DL	110001

Graduation Year	DeptID	Batch ID	CompanyID
2023	1	1	101
2023	1	2	102
2024	2	3	103
2024	3	4	104
2025	4	5	105

Current Position	Linkedin Profile	IS Active
Software Engineer	in / aaravmehra	TRUE
Data Analyst	in / priyanka	TRUE
Product Manager	in / rohangupta	TRUE
System Engineer	in / snehareddy	TRUE
Analyst	in / vikramsingh	FALSE

Normal Form Check:

Normal Form	Status	Reasoning
1NF	✓	Composite attributes (Name, Address) are decomposed into atomic components; Phone Numbers moved to Alumni-Phone; Skills and Events moved to junction relations
2NF	✓	Single-attribute primary key - no partial dependency
3NF	✓	Transitive dependencies DeptID → DeptName, BatchID → BatchYear, CompanyID → CompanyName removed by projection
BCNF	✓	Determinants AlumniID and Email are candidate keys.

Decomposition:

Step 1: 3NF : remove the transitive dependency
 $\text{Alumni ID} \rightarrow \text{Dept ID} \rightarrow \text{Dept Name}$

→ Project (Dept ID, Dept Name) out of the flat relation; it merges into department

→ (Dept ID, Dept Name) → merges into Department

Dept ID	Dept Name
1	Computer Science
2	Electronics & Comm.
3	Mechanical Engg.
4	Electrical Engg.
5	Civil Engg.

Step 2: 3NF : remove the transitive dependency
 $\text{Alumni ID} \rightarrow \text{Batch ID} \rightarrow \text{Batch Year}$

→ Project (Batch ID, Batch Year) out of the flat relation; it merges into Batch

→ (Batch ID, Batch Year) - merges into Batch

Batch ID	Batch Year
1	2023
2	2023
3	2024
4	2024
5	2025

Step 3: 3NF: remove the transitive dependency
 $\text{AlumniID} \rightarrow \text{CompanyID} \rightarrow \text{CompanyName}$

→ Project (CompanyID, CompanyName) out of the flat relation; it merges into Company.

→ (CompanyID, CompanyName) - merges into Company.

CompanyID	CompanyName
101	TCS
102	Infosys
103	Google
104	Reliance
105	Zoho

⇒ After all three projections, Alumni keeps only DeptID, BatchID, CompanyID as foreign keys → the final relation will look like table ~~mentioned~~ on previous page.

• Student

→ studentID is a single-attribute primary key ~~that is an alternative cand~~. After transitive dependency is removed, every determinant is a candidate key → the relation is in BCNF.

• Functional Dependencies:

Student_flat (StudentID, FirstName, LastName, Email, DeptID, DeptName, Enrollment Year, Current Semester, CGPA)

Student ID $\rightarrow$ { First Name, Last Name, Email, Dept ID, Enrollment Year, Current Semester, CGPA }

~~Email $\rightarrow$ Student ID (candidate key)~~
 Dept ID $\rightarrow$ Dept Name (transitive)

Candidate Key : Student ID

Student ID	First Name	Last Name	Email
1	Ananya	Joshi	ananya.joshi@gmail.com
2	Karthik	Iyer	karthik.iyer@gmail.com
3	Meera	Pillai	meera.pillai@gmail.com
4	Arijun	Desai	arijun-desai@gmail.com
5	Disha	Menon	disha.menon@gmail.com

Dept ID	Enrollment Year	Current Semester	CGPA
1	2024	4	8.9
2	2024	4	8.4
3	2025	2	7.8
4	2025	2	9.1
5	2023	7	8.2

• Normal Form Check

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single-attribute primary key $\rightarrow$ no partial dependency

3NF

0 ✓

Transitive dependency
 $\text{StudentID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$
 DeptName removed by
 projection ; DeptID as
 FK.

BCNF

✓

Determinants StudentID
 and Email are candidate
 keys

Decomposition :

Step 1 $\Rightarrow$ 3NF: remove transitive dependency $\text{StudentID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

$\Rightarrow$ Project (DeptID, DeptName) out of the flat relation;
 it merges into Department and Student
 keeps DeptID as a foreign key.

(DeptID, Dept Name) $\rightarrow$ merges into Department:

DeptID

DeptName

1

Computer Science

2

Electronic & Comm.

3

Mechanical Engg.

4

Electrical Engg.

5

Civil Engg.

$\Rightarrow$ And the table will look like on previous
 page but DeptID as foreign key

Mentorship

Functional Dependencies :

Mentorship - flat (Mentorship ID, Alumni ID, Alumni Name, Alumni Company, Student ID, Student Name, Student Dept, Start Date, End Date, Status, Mentorship Area, Goals).

Mentorship ID $\rightarrow$ { Alumni ID, Student ID, Start Date, End Date, Status, Mentorship Area, Goals }

Alumni ID $\rightarrow$ { Alumni Name, Alumni Company }

Student ID $\rightarrow$ { Student Name, Student Dept }

candidate key: (Mentorship ID, ^{Student ID, Alumni ID} surrogate PK; ~~(Alumni ID, Student ID)~~) is the owner based identification key

Mentorship ID	Alumni ID (FK)	Student ID (FK)	Start Date	End Date
1	1	1	2025-01-10	NULL
2	2	2	2025-02-01	NULL
3	3	3	2024-08-15	2025-08-30
4	4	4	2025-03-20	NULL
5	5	5	2024-07-01	2025-05-3

Status	Mentorship Area	Goals
Active	Data Science	Place in tech
Active	Career Guidance	Resume review
Completed	Web Development	Full-stack skills
Active	Research	Conference paper
Completed	Entrepreneurship	Startup guidance

Normal-Form check:

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Partial dependencies Single - attribute Alumni ID $\rightarrow$ Alumni Name, primary key $\rightarrow$ no partial Alumni Company remove key projection (step 1) and also Student ID $\rightarrow$ Student Name in step 2
3NF	✓	After 2NF decomposition no non-prime attributes depends on another non-prime attribute
BCNF	✓	The only determinant left is the composite key (Alumni ID).

Decomposition:

Step 1: 2NF: remove partial dependency

Alumni ID $\rightarrow$ Alumni Name, Alumni Company.

$\Rightarrow$ Alumni ID is a proper subset of the composite key. Project the mentor details out of the flat relation; they merge into alumni.

(Alumni ID, Alumni Name, Alumni Company) $\rightarrow$ merges into alumni

Alumni ID	Alumni Name	Alumni Company
1	Aarav Mehta	TCS
2	Periya Nair	Google
3	Rohan Gupta	Infosys
4	Ancha Reddy	Reliance
5	Vikram Singh	Zoho

Step 2: 2NF: remove partial dependency Student ID $\rightarrow$ Student Name, Student Dept

- $\Rightarrow$ Student ID is a proper subset of the composite key. Project the mental details out of the flat relation: they merge into Student.
- $\Rightarrow$ (Student ID, Student Name, Student Dept) $\rightarrow$ merge into Student.

Student ID	Student Name	Student Dept
1	Ananya Joshi	Computer Science
2	Karthik Iyer	Electronics & Comm
3	Meera Pillai	Mechanical Engg.
4	Arijun Desai	Electrical Engg.
5	Dwija Menon	Civil Engg.

- $\Rightarrow$ After this decomposition table will look like as in previous page.

• Event

→ EventID is a single-attribute primary key. The organizer detail chain is a transitive dependency removed by projection; the only determinant left is Event ID - the relation is in BCNF.

• Functional Dependency:

Event flat (EventID, Event Name, Event Type, Event Date & Venue, Organizer ID, Organizer Name)

EventID $\rightarrow$ {Event Name, Event Type, Event Date, Venue, Organizer ID}

OrganizerID $\rightarrow$ Organizer Name (BCNF violation ~~transitive~~)

Candidate Key: EventID, Attendee

EventID	EventName	EventType	EventDate
1	Alumni Reunion 2025	Reunion	2025-06-15
2	AI Workshop	Workshop	2025-04-20
3	Career Seminar	Seminar	2025-03-10
4	Startup Networking	Networking	2025-05-05
5	Freshers Meet 2025	Networking	2025-08-01

Venue	OrganizerID
Main Audi	1
Tech Hall	2
Seminar Hall	3
Innovation Lab	4
Main Audi	5

Normal Form Check:

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single attribute ^{key} primary EventID → no partial dependency, so 2NF holds without decomposition
3NF	✓	No non-prime → non-prime dependency. (Organizer ID is FK after BCNF decomposition)
BCNF	✓	Non-superkey determinant OrganizerID → Organizer Name removed by decomposition (step 1); every remaining determinant is a candidate key.

Step 1: BCNF : decompose on the non-superkey determinant OrganizerID → Organizer Name

In event-flat the key is EventID, but Organizer ID → Organizer Name holds and OrganizerID is NOT a superkey violating BCNF.

→ R1 = {OrganizerID, OrganizerName} → merges into Alumni.

Organizer ID	Organizer Name
1	Aarav Mehta
2	Brigya Nair
3	Rohan Gupta
4	Snigdha Reddy
5	Vikram Singh

→ ~~R2 = {}~~

→ The table will be like that on previous page but with organizer ID as FK.

• Donation :

Functional Dependencies :

Donation - Flat (DonationID, DonorID, Donor Name, Donor Email, Amount, Donation Date, Payment Method)

DonationID → {DonorID, Amount, Donation Date, Payment Method}

DonorID → {Donor Name, Donor Email}

Candidate Key : DonationID

DonationID	DonorID	Amount	Donation Date	Payment Method
1	1	50000	2025-01-10	Online
2	2	25000	2025-02-14	Online
3	3	100000	2025-03-01	Cheque
4	4	15000	2025-04-22	Cash
5	5	75000	2025-06-30	Online

Normal Form Cheque:

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single attribute primary key $\rightarrow$ no partial dependency
3NF	✓	Transitive dependency. $\text{DonationID} \rightarrow \text{DonorID} \rightarrow \text{DonorName}$; resolved in projection
BCNF	✓	DonationID is the only determinant and is a candidate key.

Decomposition?

Step 1) 3NF: remove the transitive dependency
 $\text{DonationID} \rightarrow \text{DonorID} \rightarrow \text{DonorName}$

$\rightarrow$ Project $(\text{DonorID}, \text{DonorName}, \text{DonorEmail})$ out of the flat relation; it merges into ALUMNI

DonorID	DonorName	DonorEmail
1	Aarav Mehta	aarav.mehta@email.com
2	Buiya Nair	buiya.nair@email.com
3	Rohan Gupta	rohangupta@email.com
4	Sneha Reddy	sneha.reddy@email.com
5	Vikram Singh	vikram-singh@email.com

→ After this and making DonorID as FK, the relation is normalized

• Job

Functional Dependency :

Job - Elat (JobID, JobTitle, JobType, Salary, PostedBy, FirstName, LastName, Email, CurrentPosition)

Job-ID → { JobTitle, JobType, Salary, PostedBy }

PostedBy → { FirstName, LastName, Email, Current Position } (transitive)

Candidate Key : Job ID

JobID	JobTitle	JobType	Salary	Posted By
1	Software Engineer	FullTime	15-20LPA	1
2	Data Analyst	FullTime	10-15LPA	2
3	Web Developer Intern	Internship	5LPA	3
4	Project Manager	Full-Time	25-30LPA	4
5	ML Engineer	Contract	20-25LPA	5

∴ Normal - Form Check ∴

Normal Form	Status	Reasoning
1NF	✓	All attributes are atomic
2NF	✓	Single attribute primary key - no partial dependency
3NF	✓	There is transitive dependency which is solved in projection
BCNF	✓	JobID is the only determinant and is a candidate key

Decomposition:

- 3NF: remove the transitive dependency
 $JobID \rightarrow PostedBy \rightarrow FirstName$
- Project (PostedBy, FirstName, LastName, Email, CurrentPosition) out of the relation
 relation $\rightarrow$ these are Alumni's own attributes, so the projection merges back into Alumni; Job keeps only PostedBy as FK.
- $PostedBy \rightarrow \{FirstName, LastName, Email, CurrentPosition\} \rightarrow$ merges into Alumni

PostedBy	FirstName	LastName	Email	CurrentPosition
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- Alumni-Skill (Junction Relation)
 Alumni-Skill (AlumniID, SkillID)
 $(AlumniID, SkillID) \rightarrow \phi$
 Candidate Keys: $(AlumniID, SkillID) \rightarrow$
 composite primary key.

Alumni ID	Skill ID
1	1
2	3
3	2
4	1
5	3

Normal-form check:

Normal Form

Status

Reasoning

1NF

✓

All values are atomic

2NF

✓

No non-key attribute exists, so a partial dependency is impossible

3NF

✓

No non-key attribute exists, so a transitive dependency is impossible

BCNF

✓

The only determinant is the full composite key, a candidate key.

Alumni-Event (Function Relation)

Functional Dependencies:

Alumni-Event (AlumniID, EventID)

(AlumniID, EventID) $\rightarrow \emptyset$

Candidate Keys: (AlumniID, SkillID) $\rightarrow$ composite primary key

AlumniID

EventID

1

1

2

2

1

1

3

3

4

4

Normal-Form check

Normal Form

Status

Reasoning

1NF

✓

All values are atomic

2NF

✓

No non-key attribute exists, so a partial dependency is impossible

3NF

✓

No non-key attribute exists, so a transitive dependency is impossible

BCNF

✓

The only determinant is the full composite key, a candidate key

⇒ Similarly with these Normalization Reasons:

Alumni-Phone

FDs: Alumni-Phone (AlumniID, Phone Number)

(AlumniID, Phone Number) → ∅

Candidate keys: (AlumniID, Phone Number)
[Composite Primary key]

AlumniID

Phone Number

1

98 400 12345

2

99 4 0012345

2

98 800 12345

3

99 000 12345

4

97 900 12345

Student-Email

FDs : Student-Email (Student ID, Email)
 (Student ID, Email) $\rightarrow \emptyset$

Candidate Keys : (Student ID, Email)
 (Composite primary key)

Student ID

Email

1

ananya.joshi@email.com

1

anany.j@email.com

2

Karthik.yeah@email.com

3

meera.hillai@email.com

4

arjun.desai@email.com

$\Rightarrow$ They Both are already in BCNF.